Syrian Private University
Faculty of artificial intelligence engineering Department of Robot engineering and intelligent

خط انتاج لتعبئة العطور

Perfume packaging production line

A quarterly project report that updated the requirements for obtaining a bachelor's degree in artificial intelligence engineering-robot engineering and intelligent systems

إعداد الطالب
محمد أنس غياث
فاكهاني

إشراف
د. سامر حسام الدين
م.شادي صعب

2025-2026

SUPERVISOR CERTIFICATION

I Certify that the Preparation of this Project entitled by ……

, prepared by ……...…

……

……., was made under my supervision at Department of Robot engineering and intelligent in partial fulfillment of the Requirements for the Degree of Bachelors of Artificial intelligence engineering-robot engineering and intelligent systems

Name: …… Signature: Date:

Abstract

Manual bottle filling in small-scale industries suffers from spillage, inconsistent fill volumes, slow operation, and high labor dependency. This project presents the design and development of a low-cost, fully automated bottle filling system controlled by an Arduino mega microcontroller. The system integrates an HC-SR04 ultrasonic sensor for non-contact bottle detection at the filling station, a YF-S201 flow sensor for real-time volumetric measurement of the dispensed liquid, a stepper motor driving a chain conveyor for precise bottle positioning, and a solenoid valve actuated through a relay to control liquid flow. A Nextion 3.5-inch HMI provides a user-friendly interface for entering the target fill volume, monitoring system status, and visualizing real-time and historical filling data through graphical charts. Unlike timer-based filling systems, the proposed design uses closed-loop volumetric feedback from the flow sensor, ensuring accurate and repeatable fill quantities regardless of pressure variations. The system operates as follows: the user enters the desired volume via the HMI, the stepper motor advances the conveyor until the ultrasonic sensor detects a bottle, the valve opens, and the Arduino counts flow-sensor pulses until the target volume is reached, after which the valve closes and the conveyor resumes, with a total system cost significantly lower than commercially available filling machines. The modular and portable design makes the system suitable for small beverage, water, and chemical bottling industries, and it can be easily adapted to different bottle sizes and volumes through software configuration only.

Keywords: Automated bottle filling, Arduino mega, ultrasonic sensor, flow sensor, stepper motor, Nextion HMI, solenoid valve, volumetric control.

.

الملخص

تعاني عمليات تعبئة القوارير اليدوية في الصناعات الصغيرة من الانسكاب، وعدم ثبات الكميات المعبأة، وبطء الإنتاج، والاعتماد الكبير على العمالة. يقدّم هذا المشروع تصميماً وتطويراً لنظام تعبئة قوارير أوتوماتيكي منخفض التكلفة يعتمد على متحكم اردوينو ميغا. يدمج النظام حساساً فوق صوتياً لكشف القارورة دون تلامس في محطة التعبئة، وحساس تدفق لقياس الحجم الفعلي للسائل المتدفق لحظياً، ومحركاً خطوياً ستيبر يقود سيراً ناقلاً من نوع السلسلة لتحديد موضع القارورة بدقة، وصماماً كهربائياً يُشغَّل عبر مرحّل للتحكم بتدفق السائل. وتوفّر شاشة نيكسشن بمقاس 3.5 بوصة واجهة استخدام سهلة لإدخال الحجم المطلوب، ومراقبة حالة النظام، وعرض البيانات اللحظية والتاريخية عبر رسوم بيانية. وخلافاً لأنظمة التعبئة المعتمدة على الزمن، يعتمد التصميم المقترح على تغذية راجعة حجمية مغلقة من حساس التدفق، مما يضمن دقة تكرار الكميات المعبأة بغض النظر عن تغيّر الضغط. وتسير عملية التشغيل كما يلي: يُدخل المستخدم الحجم المطلوب عبر الشاشة، فيحرّك المحرك الخطوي السير حتى يكشف الحساس الفوق صوتي وجود القارورة، ثم يُفتح الصمام ويبدأ اردوينو بعدّ نبضات حساس التدفق حتى بلوغ الحجم المستهدف، فيُغلق الصمام ويستأنف السير حركته ، وبتكلفة إجمالية أقل بكثير من آلات التعبئة التجارية. ويجعل التصميم المعياري والمحمول النظام مناسباً لصناعات المشروبات والمياه والكيماويات الصغيرة.

الكلمات المفتاحية: تعبئة أوتوماتيكية، ، حساس فوق صوتي، حساس تدفق، محرك خطوي، شاشة، صمام كهربائي، تحكم حجمي.

Contents

Abstract ……......II

الملخص III

Index of Figure IV
Index of Table IIV
Chapter1: Introduction 2

1.1. Overview: 3

1.2. Problem Statement: 3

1.3. Project Objectives: 3

1.4. Basic Definitions: 3

1.5. Project scope: 4

1.6. Methodology: 4

1.7. Report structure: 4

1.8. Conclusion..... 5
Chapter 2: Literature review 6

2.1. Literature preview: 7

2.2. Analysis of previous studies..... 8

2.3. Research gap 9

2.4. Theoretical frame 10
Chapter 3: Working environment 24

4.1. System Architecture and Design: 25

3.2. Mathematical model: 26
Chapter 4: Quantitative Results Analysis 34
Fifth chapter: Conclusion 37
References: 39

Index of Figure

FIGURE 1: MICROCONTROLLER AND ITS TYPES ……12
FIGURE 2: ULTRASOUND PROPAGATION ……14
FIGURE 3: ARDUINO UNO ……17
FIGURE 4: HC-SR04 ULTRASONIC ……..18
FIGURE 5: TWO-CHANNEL RELAY MODULE ……..19
FIGURE 6: PUMP ……19
FIGURE 7: BREADBOARD & WIRES ……...20
FIGURE 8: POWER SUPPLY ……..21
FIGURE 9: A4988 Motor Driver…….. ……...…….21
FIGURE 10: Limit Switch Sensor …….……...22
FIGURE 11: SYSTEM ARCHITECTURE DIAGRAM 25
FIGURE 12: CIRCUIT CONNECTION MATRIX ……..27
FIGURE 13: SOFTWARE CONTROL FLOW DIAGRAM ……28
FIGURE 14: COMPONENT INTERACTION SEQUENCE DIAGRAM ……30
FIGURE 15: POWER DISTRIBUTION AND SAFETY DIAGRAM ……31
FIGURE 16: STATE MACHINE DIAGRAM ……..32
FIGURE 17: schematic diagrame……. ……33

Index of Table

TABLE1: Comparison between existing research models and the proposed Arduino-Ultrasonic architecture……..……9
TABLE 2: TECHNICAL RESULTS TABLE ……..39
TABLE 3: COMPARISON TABLE WITH PREVIOUS STUDIES ……40

Chapter1: Introduction

1.1. Overview:

The Automated Bottle Filling System is a low-cost, microcontroller-based solution for small-scale bottling industries. It eliminates manual filling problems such as spillage, inconsistent volumes, and slow production.

The system uses an Arduino mega microcontroller to coordinate all operations. An HC-SR04 ultrasonic sensor detects bottles without contact, while a YF-S401/YF-S201 flow sensor measures the dispensed volume in real time through digital pulses, enabling accurate closed-loop volumetric control. A stepper motor drives a chain conveyor for precise bottle positioning, and a solenoid valve (via relay) controls liquid flow from the reservoir.

A Nextion 3.5-inch HMI allows the user to enter the target volume, start/stop the process, and view real-time data and graphical charts.

1.2. Problem Statement:

Manual bottle filling in small-scale industries suffers from several critical shortcomings: liquid spillage during filling, inconsistent volumes from one bottle to another, slow operation due to human limitations, and high dependency on labor. These issues lead to product waste, customer dissatisfaction, and non-compliance with declared fill volumes.

1.3. Project Objectives:

The main objectives of this project are:

- ✓ To design and develop a low-cost automated bottle filling system suitable for small-scale industries
- ✓ To achieve accurate and repeatable fill volumes
- ✓ to provide a user-friendly touch interface via a Nextion HMI that allows the operator to set the target fill volume, start/stop the process
- ✓ Reduce operational costs and labor dependence.
- ✓ To visualize filling data graphically by displaying actual versus target volumes

1.6. Conclusion

This chapter provides a general introduction outlining the overall framework of the study, including a review of the research problem, its objectives and importance, as well as the scope of the study and its methodology.

1.7. proposed solution

The proposed solution is a low-cost, fully automated bottle filling system controlled by an ESP32 microcontroller. The system replaces manual filling with an integrated sequence of sensing, actuation, and user interaction:

· An HC-SR04 ultrasonic sensor detects the presence of a bottle at the filling station without contact.
· A YF-S401/YF-S201 flow sensor measures the dispensed liquid in real time by generating digital pulses proportional to volume, enabling closed-loop volumetric control instead of timer-based filling.
· A stepper motor drives a chain conveyor to position bottles precisely under the nozzle.
· A solenoid valve, actuated through a relay, opens and closes the liquid flow from the reservoir.
· A Nextion 3.5-inch HMI allows the user to enter the target fill volume, start/stop the process, and view real-time data and graphical charts.

Chapter 2: Literature review

This chapter establishes the academic foundation for the automated perfume filling system. It reviews the core technological frameworks of industrial automation, evaluates sensor and fluid dynamics literature, provides a critical analysis of previous studies, presents a comparative matrix of existing systems, and isolates the specific research gap this project addresses.

The last part of this chapter presents the theoretical frame of this project, by presenting micro controllers, sensors and actuators that are mainly used in filling automation process.

2.1. Literature preview:

The automation of liquid filling systems has been a subject of extensive research due to its industrial significance. Several researchers have explored different approaches to enhance dosing accuracy and system reliability:

- Automatic Bottle Filling System Using Arduino

1. Objective and Motivation
The primary goal of this project is to design and implement a low-cost, automated liquid filling system for small and medium-scale industries (such as soft drink and medicine manufacturing). Traditional automated machines are often prohibitively expensive for small businesses, forcing them to rely on manual labor. This project addresses that gap by utilizing an Arduino Mega 2560 microcontroller to create an efficient, robust, and economically friendly alternative that minimizes human intervention and errors.

2. System Architecture and Components
The system is built around an Arduino Mega 2560 board, which serves as the central processing unit. The hardware configuration includes:

· Sensors: An Infrared (IR) transmitter and receiver pair are used to detect the presence and position of bottles on the conveyor. The IR sensor acts as an external interrupt, sending a signal to the microcontroller when the light beam is blocked by a bottle.
· Actuators: Two DC motors (12V, 30 rpm) are employed. Motor 1 drives the conveyor belt (moving tray), while Motor 2 powers the filling mechanism (rotary pump).
· Control and Interface: The system uses a 2-line, 16-character LCD display for user interaction (e.g., "PRESS START", "START FILLING"). Relay circuits (controlled via ULN2003) are used to interface the low-power Arduino signals with the higher-power motors and solenoid valves.
· Power Supply: A dual-transformer power supply is utilized to step down 230V AC to 18V AC, which is then rectified and filtered to provide separate operating voltages for the microcontroller and the relays, preventing overload.

3. Working Principle
The operation is fully automated through a sequential control loop:

1. Upon powering on, the LCD prompts the user to press the start button.
2. The conveyor belt (Motor 1) moves until a bottle interrupts the IR beam.
3. The microcontroller stops Motor 1, positioning the bottle directly under the filling nozzle.
4. The microcontroller activates Motor 2 (the pump) for a pre-calculated time duration (based on the user-defined volume) to dispense the liquid.
5. Once the time delay elapses, the pump stops, and the conveyor belt restarts to bring the next bottle into position.
6. This process repeats for the entire batch.

4. Key Findings and Limitations
The study confirms that the automated system successfully reduces work, time, and cost compared to manual filling. The fabricated model is accurate enough for applications where high precision is not critical. However, the authors note that the system is application-specific; it may slow down production time and reduce machine capability compared to high-end industrial fillers. Therefore, it is recommended for small-scale operations rather than high-speed, precision-demanding environments.

5. Conclusion
The project demonstrates that integrating an Arduino board with a mechanical conveyor and IR sensors can effectively automate the bottle filling process. By replacing expensive industrial controllers with a widely available microcontroller, the system significantly lowers the barrier to entry for small-scale industries, allowing them to diversify and improve their operational efficiency.

Automated Bottle Filling and Capping Plant (Pandey et al., 2025)

Citation:
Pandey, G., Pokhrel, B., Budhathoki, S., & Mandal, S. K. (2025). Automated Bottle Filling and Capping Plant. Honeynexus, 2024. Kantipur Engineering College, Tribhuvan University, Nepal.

---

1. Objective and Motivation

The study addresses the limitations of manual and semi-automated bottling processes—namely labor intensity, inconsistent fill levels, spillage, and uncapped bottles—which increase waste and operating costs. The authors aim to develop a low-cost, Industry 4.0-compliant prototype that integrates computer vision with embedded control to perform inline filling, capping, and quality inspection of carbonated liquids.

2. System Architecture

The prototype is built around a Raspberry Pi 4 (vision processing) and an Arduino UNO (real-time control), communicating via UART.

· Conveyor & Star-Wheel: Bottles are transported on a conveyor and indexed through a six-slot rotating station disc driven by a NEMA 17 stepper motor.
· Station 1 – Filling: A 12 V DC pump dispenses 250 mL, timed by the Arduino.
· Station 2 – Capping: A dual-servo (MG996R) mechanism aligns and tightens plastic caps.
· Station 3 – Inspection/Dispatch: Two USB webcams (1080p, 30 FPS) capture images for fill-level and cap-presence verification under controlled LED lighting.
· Sensors: IR proximity sensors and limit switches synchronize bottle position.

3. Vision Algorithms

Two algorithms run on the Raspberry Pi:

1. Canny Edge Detection – Identifies the liquid meniscus boundary.
2. Distance Calculation – Compares the detected liquid level against a predefined reference line (datum) to classify bottles as Underfilled, FilledOK, or Overfilled. Cap presence is verified via color thresholding and contour analysis.

Control Logic (Finite State Machine):

· Underfilled bottles are automatically rotated back for refilling.
· Overfilled bottles are diverted to a reject chute.
· Uncapped bottles trigger up to two capping retries before rejection.
· All data (fill volumes, cap status, rejects) is logged to an MQTT-based IoT dashboard.

4. Key Results

Testing was performed on 250 mL PET bottles with carbonated dark liquid (100 bottles per trial, averaged over five runs).

Metric Manual Method Automated System
Cycle time/bottle ~20 s ~14 s
Throughput ~180 bottles/h ~257 bottles/h (+30%)
Fill accuracy (±2%) ~90% ~97%
Underfill/spillage rate ~10% ~3%
Reject rate ~12% ~5%

· Fill-level detection accuracy: 97%
· Cap detection accuracy: ~100% classification in tests
· Total hardware cost: ~45,000 NRs (≈ US$340)

Table1: Comparison between existing research models and the proposed Arduino-Ultrasonic architecture.

Aspect	Automated bottle filling and capping plant 2025	Automatic bottle filling system using Arduino 2022
Main controller	Raspberry PI 4 + Arduino uno	Arduino mega 2560
sensors	IR + limit switches+ 2 usb webcams	IR sensor only
Detection technology	Image processing	Through IR
Fill-level inspection	97% accuracy	Timer- based dispensing only
capping	yes	no
Automatic correction	yes	no
Motion mechanism	Conveyor belt + six slot star wheel disc driven by nema 17 stepper	Flat belt conveyor + 4 dc motors
Pump	12v dc pump	Rotary pump + solenoid valve
User interface	IOT dashboard	Lcd display + push buttons
cost	340usd	Very low

an array of modern applications and industries, microcontrollers have quickly achieved widespread market penetration and can be found today in many technologies and gadgets. Any